

Software Design Document

The Car Dealership Inventory

Contents

The Car Dealership Inventory.....	1
1. Introduction.....	2
1.1. Product Scope	2
1.2. Requirements	2
1.3. Assumptions.....	2
2. Design	3-5
2.1. Class Diagram.....	3
2.2. Pseudo Code.....	4
2.3. Flow Charts.....	5
3. User Interface.....	6
4. Testing.....	7-13

1. Introduction

1.1. Product Scope

The purpose of this program is to help a car dealership manage its inventory using the principles of inheritance

1.2. Requirements

These are the requirements for this program.

- Must have a “Vehicle” base class with private data members: make (string), model (string), color (string), mpg (integer), msrp (double). Must also contain appropriate constructor and method(s)
- Must have a derived “Car” class with private data members: spareInTrunk (Boolean), rearWindShieldWiper (Boolean). Must also contain appropriate constructor and method(s)
- Must have a derived “Truck” class with private data members: sidePanelStorage (Boolean), autoReleaseTailgate (Boolean). Must also contain appropriate constructor and method(s).
- The Menu system should allow the user to: add a car or truck to inventory, print a list of current inventory, search/print a list of cars or trucks by make, model, or color, and print details about a specific car or truck.
- Must read the initial input to load the inventory
- Must have at least 4 test cases

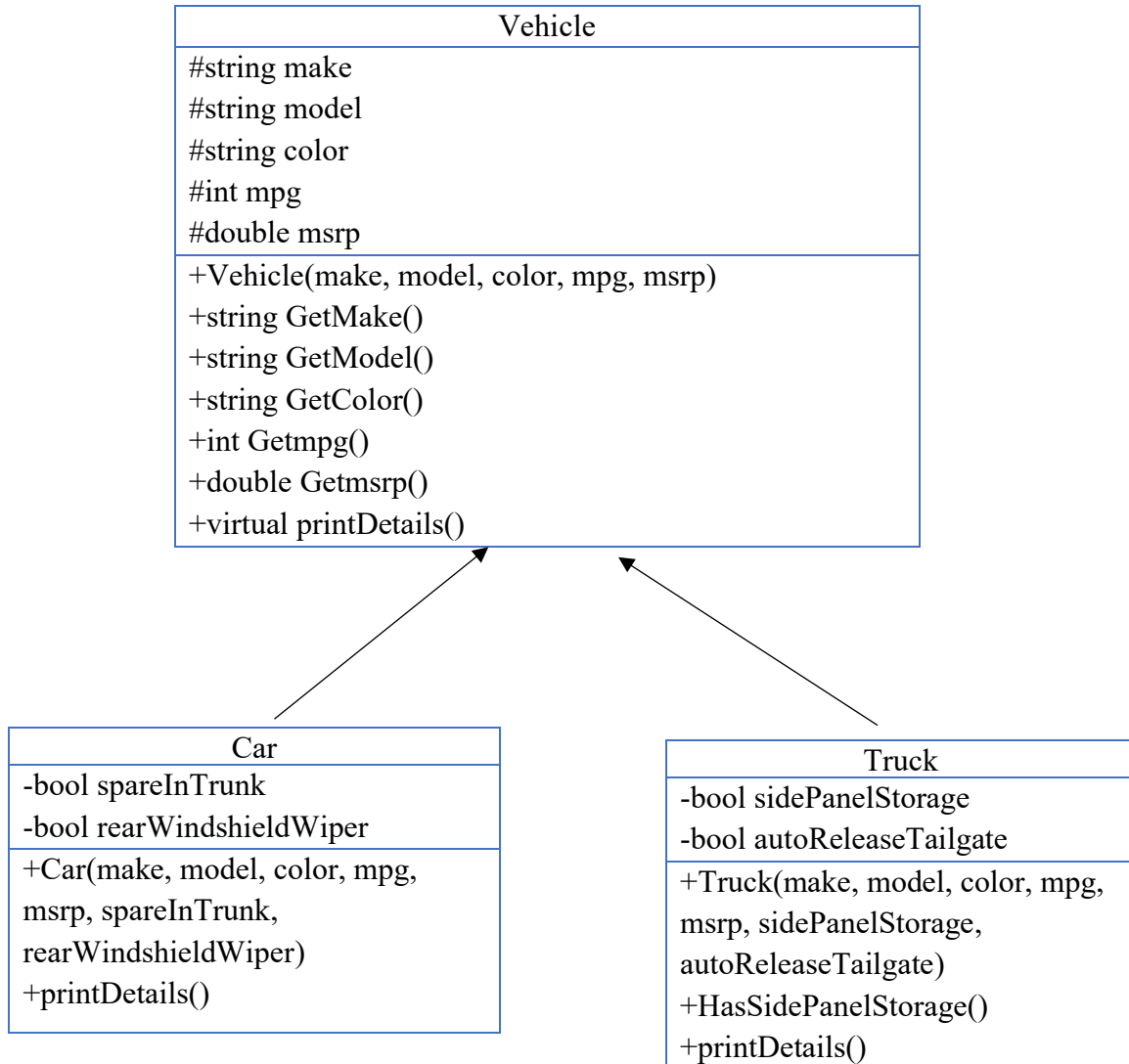
1.3. Assumptions

These are the assumptions made in the design and implementation of the program. It also covers the execution of the program.

- The user should use the menu options logically (i.e., adding a vehicle(s) THEN printing the list of the current inventory instead of printing first then adding)
- The user should only enter valid menu options; this program does not handle any menu options other than the ones provided
- We can preload our data by hard-coding the vehicle inventory
- When printing our inventory, it does not have to be in a specific order
When printing a list of vehicles by make, model, or color, the printed list can just output a specific vehicle, make, model, or color like “All red vehicles: Red [make] [model]...” and the same for if we want to do it for make or model
- In addition to the previous assumption, we can have both cars AND trucks in that list, not just strictly all cars or all trucks
- When “printing details about a specific car or truck”, it just means that we ask the user if it is a car or a truck, and the make/model/color, and then print out all of the attributes associated with the vehicle, like msrp, mpg, etc.
- By extension of the previous assumption, no vehicle with the same make, model, and color will have the same remaining attributes (like spareInTrunk or mpg)

2. Design

2.1. Class Diagram



2.2. Pseudo Code

(This pseudo code emulates the main function in main.cpp to demonstrate program functionality so that no function definitions will be implemented)

Welcome the user to the car dealership system.

Initialize a vector with pointers to both Car and Truck objects.

Hardcode our preloaded data (at least 6 vehicles: 3 cars and 3 trucks) and add them to the inventory

While the user is not done using the program (menu option 5):

- Ask the user for a menu option.

- If the menu option is 1:

- Make a call to the AddVehicle() method (defined in main.cpp)
 - Ask the user if they want to add a car or a truck
 - Ask the user for the necessary information, depending on whether it is a truck or a car
 - Add the vehicle to the vector

- If the menu option is 2:

- Make a call to the PrintList() method (defined in main.cpp) to print the list of the inventory

- If the menu option is 3:

- Make a call to the SearchPrintInventory() method (defined in main.cpp)
 - Ask the user if they wish to search by make, model, or color
 - Search and print each vehicle based on the condition (in the implementation, it will be a for loop with 3 conditionals)

- If the menu option is 4:

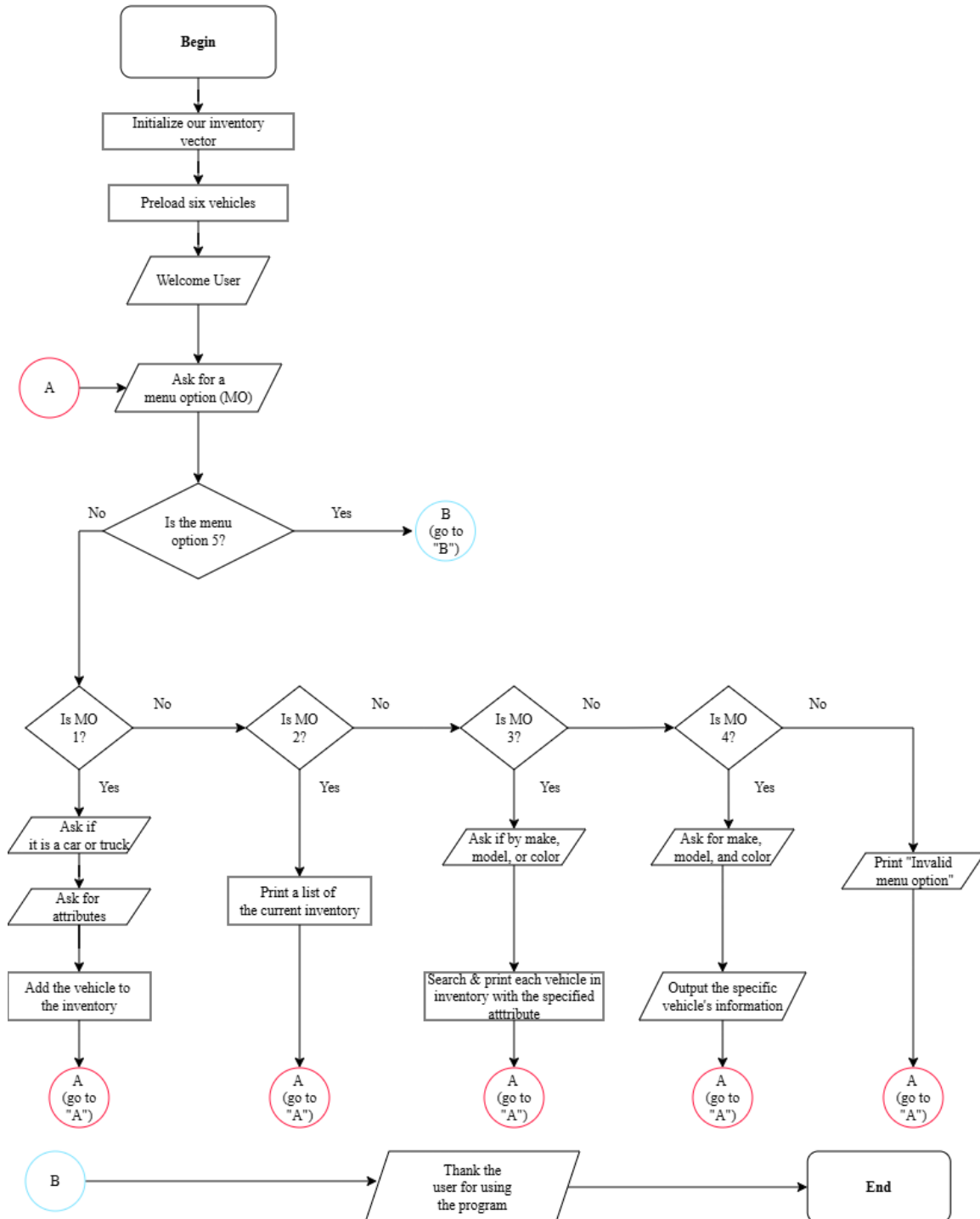
- Make a call to the PrintVehicle() method (defined in main.cpp)
 - Ask the user for the make, model, and color (this will prompt a search for the vehicle)
 - Output the vehicle's information

- If the menu option is outside the given range:

- Print "Invalid menu option."
 - Exit the loop and ask for another menu option

Thank the user for using the program.

2.3. Flow Charts



3. User Interface

This program will be run on a command line prompt. When the program starts, it displays a welcome message and a menu with five menu options that the user can perform (listed below for reference). The user should use the menu options logically and carefully (such as adding an order from the queue and then removing it).

The user may utilize the menu options as many times as they wish. Once the user enters menu option 5, the program will display a message thanking the user for using the program and thus ending it. The user may run the program as many times as they wish.

Menu Options:

1. Add a car or truck to inventory
2. Output a list of the current inventory
3. Search, and output a list of cars, or trucks, by make, model, or color
4. Output details about a specific car or truck
5. End the program

4. Testing

I will test if the program is correct and ensure that all the menu options function as intended using the following test cases.

For our purposes, some menu options may be used with other menu options in a given test case to demonstrate their functionality.

For brevity, I will not include prompts asking the user for vehicle details in the expected output, nor the output of a menu option used to demonstrate the functionality of a different menu option (unless it is needed for that demonstration).

I will display the welcome and closing messages only in the first and last test cases, respectively.

Note: For the test cases regarding the functionality of menu option 3, I have added another vehicle, but it is not required for your sake when testing

Test Cases and the preloaded inventory (for reference) are on the following pages

For reference, here is our preloaded data:

1. **Car**
Make: Toyota
Model: Camry
Color: Silver
MPG: 32
MSRP: 27495.50
HasSpareInTrunk: Yes
HasRearWindshieldWiper: No
2. **Truck**
Make: Ford
Model: F-150
Color: Black
MPG: 20
MSRP: 41560.75
HasSidePanelStorage: Yes
HasAutoReleaseTailGate: No
3. **Car**
Make: Honda
Model: Accord
Color: Blue
MPG: 30
MSRP: 28940.60
HasSpareInTrunk: Yes
HasRearWindshieldWiper: No
4. **Truck**
Make: Chevrolet
Model: Silverado 1500
Color: Red
MPG: 19
MSRP: 43620.85
HasSidePanelStorage: No
HasAutoReleaseTailGate: Yes
5. **Car**
Make: Hyundai
Model: Elantra
Color: White
MPG: 36
MSRP: 22325.40
HasSpareInTrunk: No
HasRearWindshieldWiper: Yes
6. **Truck**
Make: Toyota
Model: Tacoma
Color: Gray
MPG: 21
MSRP: 34980.95
HasSidePanelStorage: Yes
HasAutoReleaseTailGate: Yes

Test Case 1

Purpose:

This is to test the functionality of the menu options 1 and 2.

Inputs:

1
Car
Nissan
Altima
Gray
31
28100.45
Yes
No
2

Expected Output:

Welcome to our car dealership's inventory management system! Please select one of the following menu options below!

Done. The vehicle has been added!

Current Inventory

Silver Toyota Camry (Car) with 32 MPG and \$27495.50 MSRP

-Spare in the trunk: Yes

-Has rear windshield wiper: No

Black Ford F-150 (Truck) with 20 MPG and \$41560.75 MSRP

-Has side panel storage: Yes

-Has auto-release tail gate: No

Blue Honda Accord (Car) with 30 MPG and \$28940.60 MSRP

-Spare in the trunk: Yes

-Has rear windshield wiper: No

Red Chevrolet Silverado 1500 (Truck) with 19 MPG and \$43620.85 MSRP

-Has side panel storage: No

-Has auto-release tail gate: Yes

White Hyundai Elantra (Car) with 36 MPG and \$22325.40 MSRP

-Spare in the trunk: No

-Has rear windshield wiper: Yes

Gray Toyota Tacoma (Truck) with 21 MPG and \$34980.95 MSRP
-Has side panel storage: Yes
-Has auto-release tail gate: Yes

Gray Nissan Altima (Car) with 31 MPG and \$28100.45 MSRP
-Spare in the trunk: Yes
-Has rear windshield wiper: No

Test Case 2

Purpose:

This is to test the functionality of menu option 3 by make.

Inputs:

1
Car
Toyota
Corolla
Green
35
24120.75
Yes
No
3
1 (*this indicates that we want to search by make*)
Toyota

Expected Output:

The vehicles in the inventory with the make “Toyota” are:
-Silver Toyota Camry
-Gray Toyota Tacoma
-Green Toyota Corolla

Test Case 3

Purpose:

This is to test the functionality of the menu option 3 by model

Inputs:

3
2
F-150

Expected Output:

The vehicles in the inventory with the model “F-150” are:
-Black Ford F-150

Test Case 4

Purpose:

This is to test the functionality of menu option 3 by color

Inputs:

1
Truck
Ford
Ranger
White
22
38210.65
No
Yes
3
3
White

Expected Output:

The vehicles in the inventory with the color “White” are:
-White Hyundai Elantra
-White Ford Ranger

Test Case 5

Purpose:

This is to test the functionality of menu option 3 if nothing is found.

Inputs:

3
3
Maroon

Expected Output:

Sorry! No such vehicle was found with your specified attribute.

Test Case 6

Purpose:

This is to test the functionality of menu option 4

Inputs:

4
Honda
Accord
Blue

Expected Output:

VEHICLE INFORMATION:
Blue Honda Accord (Car) with 30 MPG and \$28940.60 MSRP
-Spare in the trunk: Yes
-Has rear windshield wiper: No

Test Case 7

Purpose:

This is to test the functionality of menu option 4 if there is no vehicle found.

Inputs:

4
Ford
Tacoma
Blue

Expected Output:

Sorry! Unfortunately, no such vehicle exists in our inventory.

Test Case 8

Purpose:

This is to test if there is an invalid menu option received.

Inputs:

10

Expected Output:

Sorry! Invalid menu option.

Test Case 9

Purpose:

This is to test the functionality of menu option 5

Inputs:

1

Car

Honda

Civic

Blue

34

25240.80

Yes

No

5

Expected Output:

Done. The vehicle has been added!

Thank you for using our car inventory system! Have an amazing day!